Aerosols in the vacuum: modelling the time variability of the plume of Enceladus

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Key Points:

- A one dimensional model couples liquid level gas flow and wall heat diffusion in a tidally flexing south polar crack on Enceladus
- Diurnal widening and narrowing of the crack drives the water level up and down and modulates the water vapor mass flux of the plume
- Heat stored in a thin wall layer produces a secondary plume peak near mean anomaly fifty degrees consistent with observations

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Abstract

Previous studies have examined the dynamics of water vapor arising from liquid water in the cracks near the south pole of Enceladus (Nimmo, Spencer, et al., 2007; Schmidt, Brilliantov, et al., 2008; Ingersoll & Pankine, 2010; Postberg, Schmidt, et al., 2011; Nimmo, Porco, et al., 2014; Kite & Rubin, 2016; Nakajima & Ingersoll, 2016). The observed plume activity has a main brightness peak near mean anomaly (MA) of 180 degrees and a weaker secondary peak near MA of 50 degrees (Hedman, Gosmeyer, et al., 2013; Ingersoll, Ewald, et al., 2020; Nimmo, Porco, et al., 2014), the latter present only for narrow cracks. We show that this secondary peak is not a fine-tuned coincidence but an essentially unavoidable consequence of an active crack. In our model the water level rises and falls with the diurnal widening and narrowing of the walls, and the rising plume continually condenses ice back onto the walls and seals the crack; because the water level is highly sensitive to the gap, any sufficiently flexed crack — regardless of its initial width — is driven within a few diurnal cycles to a narrow, near-overflow state in which the water periodically reaches the surface and produces the secondary peak. The absolute tidal swing of the walls, rather than acting as a threshold, sets only how far the crack must seal first. We predict the phase and relative strength of the resulting two mass-flux peaks, reproducing the observed MA 180/50 structure and a cycle-mean emission of order the observed 200–400 kg/s. The rising water keeps the vent open against the rapid condensation, sustaining a two-way exchange of water between the oxidizing surface and the reducing ocean that is relevant to the habitability of Enceladus (Parkinson, Liang, et al., 2008).

Plain Language Summary

Enceladus, an icy moon of Saturn, sprays jets of water vapor and ice grains into space from warm cracks near its south pole. The brightness of these plumes rises and falls over each 33-hour orbit, with a strong peak and a weaker secondary peak at a specific point in the orbit. We build a simple physical model of a single crack whose walls are squeezed and stretched by Saturn's tides, causing liquid water inside to rise and fall like a piston. The narrower the crack, the higher the water is driven. We find that the escaping vapor continually freezes back onto the walls and narrows the crack, and that this self-sealing acts as a trap: whatever its starting width, any crack that flexes with the tides is squeezed within days to a narrow state in which the water reaches the surface and erupts, producing the secondary brightening. The rising water keeps the vent from freezing shut, sustaining an exchange of water between the surface and the buried ocean that matters for the moon's potential to support life.

1 Introduction

In 2005, the Cassini spacecraft found a geologically active south pole on Saturn's moon Enceladus (Porco, Helfenstein, et al., 2006). Plumes that contain vapor and icy particles were identified and were considered to supply Saturn's E-ring. The plumes emanate from four major fractures that sum up to about 500 km in length, and their locations agree well with the hotspots identified in infrared images (Spitale & Porco, 2007). The analysis of the Cassini Ultraviolet Imaging Spectrograph (UVIS) data in solar occultation reveals that the total rate of emission of water vapor from the surface is 200 400 kg/s (Hansen, Shemansky, et al., 2011; Hansen, Esposito, et al., 2017). Data from the Cosmic Dust Analyzer (CDA) and the Ion and Neutral Mass Spectrometer (INMS) revealed macromolecular organic compounds in the icy grains (Postberg, Khawaja, et al., 2018). Being a region with far more infrared emission than other parts of the planet, the value of the non-solar radiated power from the south polar region ranges from 4.2 GW (Spencer, Howett, et al., 2013) and 5.8±1.9GW (Spencer, Pearl, et al., 2006) to 15.8±3.1GW (Howett, Spencer, et al., 2011).

By observing the details of the plume structures and compositions, the ice/vapor ratio is estimated to be 0.35 0.70, indicating that a liquid reservoir might be the source of the plumes (Ingersoll & Ewald, 2011). Further studies that reveal the existence of salt-rich ice grains and organic compounds in the plumes increased the confidence that the plumes originated from a liquid source (Postberg, Khawaja, et al., 2018). It has been confirmed later that the amplitude of the forced libration of Enceladus is too large for a model in which the icy crust is rigidly connected with its rocky core (Thomas, Tajeddine, et al., 2016). This implies the presence of a global ocean on Enceladus. If the tiger stripes are considered strike-slip faults, the shear heating would be sufficient to support the existence of liquid water in the cracks (Parkinson, Liang, et al., 2008). If two walls are decoupled and form a water-filled crack, the turbulent dissipation could supply the observed amount of heat emission from the south pole (Kite & Rubin, 2016).

Our goal is to further understand the dynamics and time variability of the plumes, as shown in Fig. 1. The plume brightness, which is related to the abundance of micron-sized particles, is 4-5 times greater at the apocenter than at the pericenter. The maximum occurs about 20 degrees in orbital phase after the apocenter. A secondary maximum occurs about 40 degrees after the pericenter (Hedman, Gosmeyer, et al., 2013; Ingersoll & Ewald, 2017). The variability is thought to be evidence of the opening and closing of the cracks due to the eccentricity tide, potentially with a physical libration and a viscoelastic interior (Hurford, Helfenstein, et al., 2007; Nimmo, Porco, et al., 2014). The timing of the plume eruptions has also received explanations from a specific interior viscosity structure (Běhounková, Souček, et al., 2017).

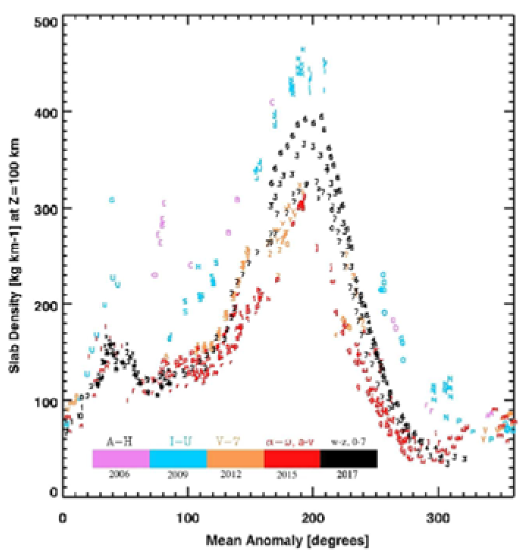

Figure 1. Slab density vs mean anomaly, adapted from Ingersoll, Ewald, et al. (2020). The data are grouped in three-year intervals and show a distinct secondary peak at around MA 50 degrees. Slab density is the mass of ice particles per unit thickness of a horizontally infinite slab above the south pole. The figure shows slab density at 100 km altitude. It is derived from the observed brightness using a model of how the particles scatter light.

In this paper, rather than seeking a particular crack configuration that reproduces the secondary peak observed in Fig. 1 (Hedman, Gosmeyer, et al., 2013; Ingersoll, Ewald, et al., 2020; Nimmo, Porco, et al., 2014), we ask whether it must occur at all. We find that condensation of the rising plume back onto the walls continually seals the crack, driving any sufficiently flexed crack — whatever its initial width — toward a narrow, near-overflow state in which the water periodically reaches the surface. The secondary peak is then a generic consequence of an active crack rather than an outcome we tune for; our aim becomes to explain why it is essentially unavoidable, to predict its phase and strength, and to use the rise and fall of the water as an indicator of two-way mass transfer between the surface and the ocean. We present the configuration of the model and the simulation results and compare them with the observations.

The thermal coupling between the liquid water and the walls was previously modeled by considering the turbulent dissipation of water as it moved up and down in response to the widening and narrowing of the cracks (Kite & Rubin, 2016). Changes in the height of the liquid-gas interface were noted but not included in their discussion of heat dissipation. We show that water level changes are important only when the mass of the plumes is considered. Such rising and falling of the water level come from vertical pressure gradients, one due to the inertia of the fluid and the other due to friction with the walls.

2 Methods

In this work, we use a one-dimensional model to study how the mass flux changes during one period of Enceladus' rotation. The novelty of our model is the rising and falling of the water level, together with its associated effects. Nakajima and Ingersoll (2016) suggested a controlled boiling model, which describes a mechanism of the plume evaporating from the gas-liquid interface held near 273 K and the fluid dynamics of the gas and liquid in the crack (regions 2 and 4 in Fig. 2). Our work also solves the time-dependent heat equation for a thin layer (region 5) at the walls: As the water rises and the wall is covered by water, it absorbs heat by conduction, and when the water level falls and the wall is exposed, it releases heat by sublimation. The time-averaged net heat due to condensation on the walls is conducted through the ice in region 3 and ultimately radiated to space in region 1.

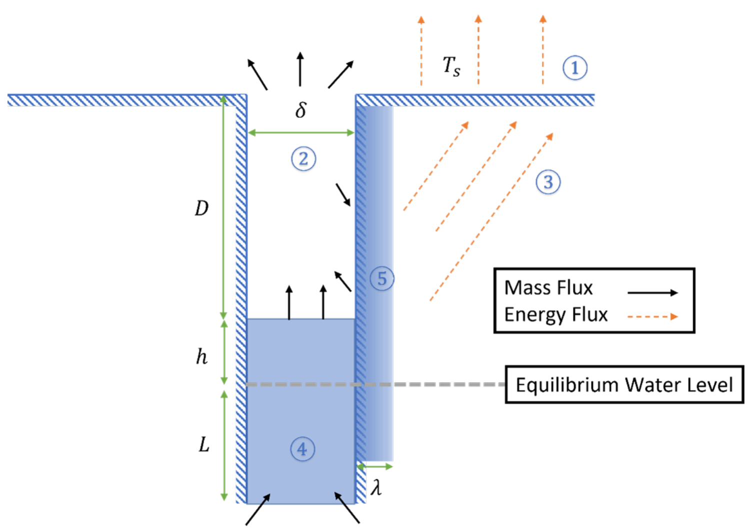

Figure 2. Illustration of the problem. Different regions are denoted by numbers in the circles, while the interactions between regions are denoted by arrows.

We assume a sinusoidal time dependence of crack width, with a minimum at the phase of 0 degrees and a maximum at the phase of 180 degrees, as in Equation 1.

$$\delta(t) = \delta_{min}\left(\frac{R_\delta + 1}{2} - \frac{R_\delta - 1}{2}\cos\frac{2\pi t}{P}\right), \tag{1}$$

Here t is the time where the model phase is at 0 degrees, $\delta(t)$ is the crack width, δ_{min} is the smallest width, and R_δ is the ratio of the maximum width to the smallest. P is 33 hours, the period of Enceladus' rotation and revolution, being the same as it is tidally locked. Equivalently, the wall motion is characterized by its absolute swing $\Delta\delta = \delta_{max} - \delta_{min} = (R_\delta - 1)\delta_{min}$, which is the quantity set by the tidal stress and is conserved as the crack seals (Section 3.2).

The single cosine of Equation 1 captures the dominant diurnal term, but the tidal stress that flexes the tiger-stripe cracks is not a pure sinusoid at the orbital frequency: orbital eccentricity, physical libration, and the shell's elastic response contribute higher harmonics (Nimmo, Porco, et al., 2014). For the diurnal fit of Section 3.5 we therefore admit a second (twice-orbital, $2f$) harmonic, replacing the profile of Equation 1 by a double-cosine

$$\delta(t)\ \propto\ -\tfrac{2}{3}\cos\tfrac{2\pi t}{P}\ +\ \tfrac{\alpha}{6}\cos\Big(2\big(\tfrac{2\pi t}{P} - \phi_2\big)\Big),$$

rescaled to the same minimum and maximum width so the absolute swing $\Delta\delta$ — and hence the self-sealing attractor of Section 3.2 — is unchanged; only the intra-cycle shape differs. The second-harmonic amplitude α and phase ϕ_2 are free parameters of the fit, and $\alpha = 0$ recovers Equation 1 exactly.

[PLACEHOLDER] *Add a dedicated reference for the second-harmonic (dou cosine) tidal-forcing law if one is available beyond the general multi-harmonic stress treatment.*

In Fig. 1, L is the equilibrium depth of the liquid water column, h is the water level height above the neutral water level, D is the instantaneous depth of the water level, and the height z is set to be zero at the bottom of the crack (where it connects to an ocean) and increases upwards. With the walls widening and narrowing, the liquid water moves vertically, and its velocity is a function of both time and height, $v = v(z,t)$. For convenience, we denote the velocities at the entrance of the crack and the gas-liquid interface as $v_0 = v(0,t)$ and $v_h = v(h+L,t)$, respectively.

2.1 Fluid dynamics of the liquid

In this part of the model, the two dependent variables are the velocity of the water v_0 at the entrance of the crack and the water level height h. These variables are functions of time and are solved from equations of continuity and momentum. Inertial, gravitational, and frictional forces are applied to the liquid water column.

The velocities at other locations are calculated from continuity by

$$v(z,t) = v_0(t) - \frac{1}{\delta}\frac{d\delta}{dt}z, 0 \le z \le h + L. \tag{2}$$

The time derivative of the entrance velocity is calculated from the momentum equation

$$\rho_w \delta \frac{\partial v}{\partial t} = -\rho_w \delta v \frac{\partial v}{\partial z} - \delta\frac{\partial p}{\partial z} - \delta\rho_w g - f\hat{v}, \tag{3}$$

Here $g = 0.113\ \mathrm{m^2 s^{-1}}$ is the gravitational acceleration, f is the magnitude of the friction force, and $\hat{v}$ is the sign of the velocity. The time derivative of the entrance velocity at the base of the crack is calculated by substituting Equation 2 into Equation 3 and integrating from $z = 0$ to $z = h + L$, assuming that the bottom of the crack has a fixed hydrostatic pressure of gL:

$$\frac{dv_0}{dt} = \frac{1}{h+L}\left[-\frac{1}{2}\left(v_h^2 - v_0^2\right) - \frac{1}{2}\left[\left(\frac{d\delta}{dt}\right)^2\frac{1}{\delta^2} - \frac{d^2\delta}{dt^2}\frac{1}{\delta}\right](h+L)^2 - gh - \int_0^{h+L}\frac{2C_f}{\delta}v^2\hat{v}\,dz\right]. \tag{4}$$

Here v_h is the velocity at the top of the liquid, or the velocity at $z = h+L$. The first two terms on the right-hand side of Equation 4 are from the advection term in Equation 3; the third term is from the gravitational force; and the last term is from the friction force with the two walls. The magnitude of the frictional stress on each wall is calculated as

$$f = C_f\rho_w v^2, \tag{5}$$

where $\rho_w = 1000$ kg/m^3 is the density of liquid water. The Fanning friction coefficient C_f is not held fixed but is evaluated at the local Reynolds number $Re = \rho_w|v|D_h/\mu$ (hydraulic diameter $D_h = 2\delta$, dynamic viscosity $\mu = 1.8\times10^{-3}$ Pa s) using the single-equation correlation of Churchill (1977), which blends the laminar limit $C_f = C_{\rm lam}/(8\,Re)$ (with $C_{\rm lam} = 96$ for an infinite parallel-plate gap) and the turbulent regime smoothly across all flow regimes (following Ingersoll and Pankine (2010); Arya (2001)). The appearance of $\hat{v}$ in Equations 3 and 4 shows that the direction of the friction force is opposite to the local velocity direction as given in Equation 1. For the smallest width considered in this work, the effect of surface tension is 4 orders of magnitude smaller than the gravitational effect of a 1-meter thick water column and hence is not considered in Equation 4.

The time derivative of h is the velocity at location $h+L$ and is obtained from Equation 2

$$\frac{dh}{dt} = v(h+L,t) = v_0 - \frac{h+L}{\delta}\frac{d\delta}{dt}. \tag{6}$$

Equations 4 and 6 are the non-linear, time-dependent equations for the dependent variables $v_0(t)$ and $h(t)$. Detailed derivations can be found in Supporting Information.

2.2 Heat storage and evaporation

The heat transfer in the thin layer of the wall (region 5) is controlled by two processes: heat conduction (region 3) and sublimation/condensation of the water vapor (region 2). The two processes differ greatly in their characteristic time scale. Region 5 serves as a thermal conduction layer that connects the two parts of the model. It has a thickness of 1 meter, which is several times the thermal skin depth corresponding to the time of a diurnal cycle ($\sim$0.33 m). This thickness allows us to resolve most of the variabilities associated with a diurnal cycle.

A section of the wall is covered for part of the diurnal cycle, and the wall absorbs heat from the water when it is submerged. When the water level falls, these sections of the wall are exposed and the water vapor either evaporates from or condenses onto the exposed walls. The sublimation rate (negative if condensation) per unit area of an exposed wall is calculated as

$$E(z,t) = \frac{p_w(z,t) - p_g(t)}{\sqrt{2\pi R T_w(z,t)}}, \tag{7}$$

Where $p_g(t)$ is the pressure of the gas at region 2 at time t, $p_w(z,t)$ is the saturated vapor pressure for the wall at height z and time t, and $T_w(z,t)$ is the temperature of the wall at height z and time t.

We name the two boundaries of the thermal layer (region 5) as the sublimation-side boundary (interface with regions 2 and 4) and conduction-side boundary (interface with region 3), respectively. The evaporation-side boundary of the shallow layer is set to have a temperature gradient pertinent to the rate of condensation or sublimation at the wall surface.

The saturated vapor pressure is obtained to a good approximation from $p_w = A\exp\left(-B/T_w\right)$ where $A = 3.63 \times 10^{12}$ Pa and $B = 6147$ K (Nakajima & Ingersoll, 2016).

At the surface (region 1), the heat release is controlled by blackbody radiation, and within the ice crust (region 3) steady-state heat conduction happens (Ingersoll & Pankine, 2010), as in Equation 8. Here $T_s(d)$ is the temperature of the surface with a horizontal distance d from the crack, and λ is the thickness of the thermal layer, or region 5. The conduction-side boundary of the thermal layer is set to match the steady-state heat conduction in region 3 at the boundary with region 5.

$$F_c = 4k\frac{T_s(d) - T(z, \lambda, t)}{\pi d}. \tag{8}$$

Eaccrack has two walls, one on either side. Details of the heat transfer are shown in Supporting Information.

2.3 Fluid dynamics of the gas

The gas dynamics (region 2) are described by a model that is similar to that in Nakajima and Ingersoll (2016). However, the sublimation and condensation happening on the surface of the walls (region 5) are treated differently, as we are now solving a transient state. Similar to Nakajima and Ingersoll (2016), the flow is initialized with evaporation at the liquid surface and travels upwards. Besides the thermodynamic variables pressure and temperature, and the vertical velocity of the gas, we keep track of the Mach number by integrating from the gas-liquid interface to the exit of the crack:

$$\frac{dM}{M}(1 - M^2)\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1} = \left(\frac{E}{\rho v \delta} + \frac{\gamma C_f}{2\delta}M^2\right)dz. \tag{9}$$

Noticeably, the Mach number condition $M = 1$ at the exit is the physical limit of the compressible flow in a frictional duct, which is the physical constraint used in Nakajima and Ingersoll (2016). However, from Equation 9, we note that in the presence of mass removal ($E < 0$), the Mach number is allowed to exceed 1. Hence, we use a different exit boundary condition here.

According to Dong, Hill, et al. (2011), the best-fit exit Mach number of the plume is 1.4-1.8. The slight difference could be due to the expansion and acceleration near the exit of the crack. Also, we confirmed with simulations that the boundary mach number has a minimal effect on the mass flux in the range of 1.4-1.8. Hence, we use $M = 1.6$ here as the exit condition. Also, most of the conclusions in Nakajima and Ingersoll (2016) still hold: having Mach numbers of 1.4 or 1.8 as the exit condition creates a total mass flux difference of no more than 2% for shell thickness of 5 km and 5% for shell thickness of 20 km. The details of the derivation of Equation 9 can be found in Supporting Information.

Different parts are assembled into a full model. The rising of falling of water level is solved first, which is used for calculation of the mass flux. To model the variability of the mass flux, we need to solve for the flow of water vapor. The water level is evaporative, creating a flow in the crack, but the mass flux and the speed of the initiated flow depends on the initial water vapor back pressure p_b just above the water level. The flow accelerates and exchanges mass with the walls, and the value of p_b is iteratively selected to match the condition that $M = 1.6$ at the exit. The model is run for about 100 days on Enceladus until the diurnal oscillations in the thermal layer stabilize.

When the crack is sealed enough that the water reaches the surface, the excess liquid expelled by the squeezing walls overflows the lip. Exposed to the vacuum, a fraction $f_{\text{evap}} = L_f/(L_v + L_f) \approx 0.12$ of this water flash-evaporates and joins the plume while the remainder freezes, releasing the latent heat that drives the evaporation. This overflow water does not evaporate instantaneously at the lip, however: it pools in a thin surface reservoir and is released over a finite residence time $\tau \sim \rho_w \delta_{\text{pond}}/j_{\text{net}}$, where j_{net} is the Hertz–Knudsen evaporation flux into the plume back-pressure (well below the vacuum limit). We model the reservoir as a first-order buffer, $dM/dt = Q_{\text{of}}(t) - M/\tau$ with vapor release M/τ, which is mass-conserving and, being much shorter than the diurnal period, leaves the cycle-integrated overflow unchanged while smoothing its release; the resulting flux is insensitive to τ over the plausible range (0.1–1 h). In any case the overflow contributes only a small fraction of the total emission (Section 3.3).

3 Results

The controlling parameters of the model are the equilibrium water-column depth L and the absolute tidal swing of the walls $\Delta\delta = \delta_{max} - \delta_{min}$; their ranges are listed in Table 1. We deliberately span an initial minimum width from centimetres (Nakajima & Ingersoll, 2016) to a metre (Kite & Rubin, 2016), but the initial width is not a true control parameter: as shown below, the self-sealing feedback drives cracks of any initial width to a common, small effective width w_{eff}, so the observable behavior is set by $\Delta\delta$ and L alone.

Parameter	**Range explored**
Equilibrium water depth L	1.5 – 20 km
Absolute tidal swing $\Delta\delta = \delta_{max} - \delta_{min}$	2 – 40 mm
Initial minimum width δ_{min}	1 mm – 1 m (converges; see Section 3.2)
Sealed effective width w_{eff}	$\sim$1 – 10 mm (emergent, not prescribed)

Table 1. Model parameters. Only L and the tidal swing $\Delta\delta$ are true controls; the initial width is swept only to show that it is irrelevant, and the sealed width w_{eff} emerges from the self-sealing feedback.

We study how the water level responds to the wall motion; this response is what the self-sealing feedback of Section 3.2 acts upon.

3.1 Water level variation

The diurnal cycle decomposes into an expanding phase (MA 0–180°) and a squeezing phase (MA 180–360°). When the walls expand, the water level first drops by continuity and is then refilled by gravity; when the walls squeeze, the water is driven upward and falls back once the level is high. The minimum width δ_{min} sets the amplitude of this excursion through friction: a narrower crack forces the water faster and traps more of it, producing a larger rise.

This sensitivity can be captured analytically. The in-crack velocity scales as $(1/\delta)(d\delta/dt)$ and the wall friction as $(1/\delta)^2(d\delta/dt)^2$; with $d\delta/dt \ \propto \ \delta R_\delta$, the maximum water-level rise scales as

$$h_{max} \propto R_\delta^2\, \delta^{-1}. \tag{10}$$

A narrower crack and a larger width ratio therefore both drive the water closer to the surface, and for a small enough width the water reaches the surface and overflows. Because the water level depends on the (effective) width in this way, any process that systematically narrows the crack will systematically push the water toward the surface — the basis of the self-sealing feedback developed next.

3.2 Self-sealing of the crack: an attractor toward overflow

A persistent plume continuously deposits some of its rising vapor back onto the walls. Integrating the wall condensation rate from the heat-storage model (Section 2.2) over the diurnal cycle, we find that the deposition is concentrated within a thermal skin depth ($\sim$0.3 m) of the surface, where the wall is warmest, and reaches $\sim$14 mm per cycle per wall. Taken literally this would build a thin ice plug at the very lip of the crack; under the diurnal flexing such a plug cannot be supported and instead fractures and is redistributed downward by viscoelastic creep of the ice (treated here as an effective narrowing of the channel; a detailed treatment of the ice rheology is left for future work).

The consequence of a narrower effective channel is that, by continuity, the water is squeezed faster during the closing phase and rises higher. The water level responds only to the current effective width and not to the crack's original size, so cracks of very different initial widths converge on the same behavior as they seal. The relevant invariant during sealing is the *absolute* tidal swing of the walls, $\Delta w = w_{\max} - w_{\min}$, which is set by the tidal stress and the rock mechanics and is unchanged by the ice lining: the lining lowers the mean width but not the swing, so the effective width ratio $R_{\delta,\text{eff}} = (w_{\max} - 2e)/(w_{\min} - 2e)$ grows without bound as the gap closes. Since the deposition rate is essentially independent of the gap, even decimetre-scale gaps are sealed to millimetre scale within a few diurnal cycles (days to weeks).

With Δw held fixed, the water rise grows steeply as the gap closes — in the large-ratio limit $h_{max} \propto \Delta w^2\, w_{\text{eff}}^{-3}$ (combining Equation 10 with $R_\delta \simeq \Delta w / w_{\text{eff}}$) — so *every* flexing crack reaches the surface once it has sealed far enough (Figure 3a). There is no forcing threshold for the secondary peak to exist: the absolute swing sets only *how far* the crack must seal before overflowing. This seal depth w^*_{eff} (Figure 3b) is of order millimetres and depends only weakly on the swing: empirically $w^*_{\text{eff}} \propto \Delta w^{0.3}$ over $\Delta w =$ 1.5–30 mm, sub-linear and below the $\Delta w^{2/3}$ of the large-ratio limit because the effective ratio at overflow remains modest ($R_{\delta,\text{eff}} \approx$ 1.4–3.9); it is also robust across equilibrium depths of 2–20 km. The secondary-peak, near-overflow configuration is thus an *attractor* for any active crack rather than a fine-tuned corner of parameter space. Its persistence requires the rising water to keep the vent open against the rapid condensation; conversely, a very weak swing seals the crack to so small a residual gap that the vapor throughput — and hence the plume — becomes negligible.

This convergence is fast and independent of where the crack starts. Figure 4 follows the closeness of approach $h_{\max}/D$ as condensation seals the crack, starting from initial widths spanning an order of magnitude (1 cm to 10 cm). At the model's condensation supply the effective width narrows steadily, and every case reaches overflow within a handful of diurnal cycles (a wider initial gap simply takes a few cycles longer); the trajectories all collapse onto the same approach curve because the water level depends only on the current effective width. The overflow state is thus reached from any starting configuration on a timescale of days to weeks.

The deposition that drives this sealing is, however, far from uniform. Figure 5 shows the instantaneous condensation rate along the exposed wall at three phases of the cycle. At every phase the deposition is sharply concentrated within the thermal skin depth ($\sim$decimetre) of the crack mouth, where the wall is warmest, and it falls off by two to three orders of magnitude deeper down. What changes over the cycle is the *extent* of the exposed wall: when the water level is neutral or low (widening phase) the vapor sees the full gas column, from the mouth down to hundreds of metres, whereas when the water has risen to near the surface (narrowing phase) condensation is confined to the topmost few metres. The sealing is therefore both spatially concentrated at the lip and strongly

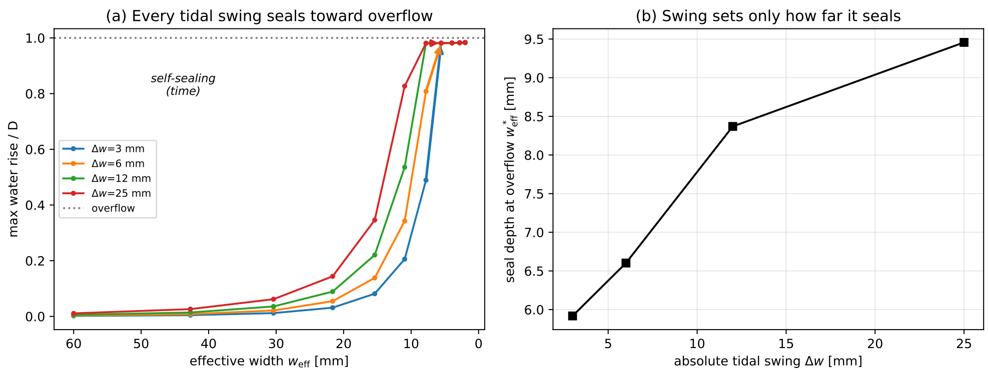

Figure 3. Self-sealing of the crack by wall condensation (effective-width model, equilibrium depth 5 km, surface $D = 500$ m above the equilibrium water level). For a fixed absolute tidal swing $\Delta w = w_{\max} - w_{\min}$ of the walls, the ice lining lowers the mean width, i.e. the effective width $w_{\rm eff}$. (a) The maximum water rise (as a fraction of D) reaches overflow for every swing Δw once the crack has sealed far enough — there is no forcing threshold. Arrows show the direction of evolution: as condensation seals the crack, $w_{\rm eff}$ decreases and the water climbs toward the surface, so the system is drawn along each curve to the overflow line (the attractor). (b) The swing sets only the seal depth $w^*_{\rm eff}$ at which overflow first occurs; it is of order millimetres and increases only weakly with the swing ($w^*_{\rm eff} \propto \Delta w^{0.3}$).

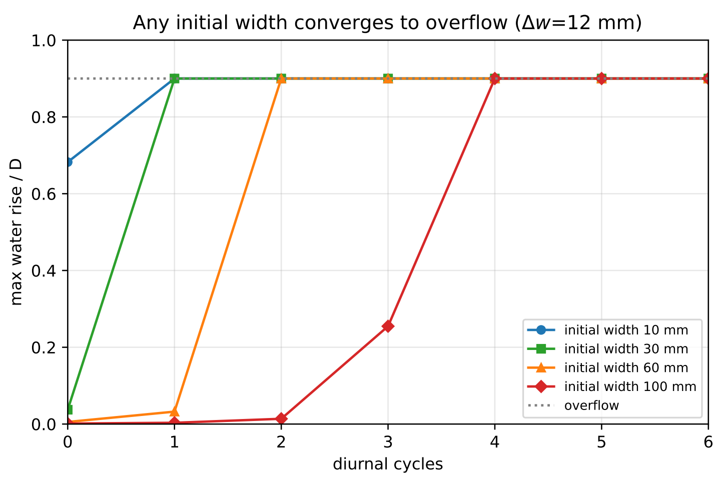

Figure 4. Convergence to the overflow attractor (equilibrium depth 5 km, $\Delta w = 12$ mm). Each curve follows the maximum water rise $h_{\max}/D$ as wall condensation narrows the effective width, cycle by cycle, from a different initial width (1–10 cm). Regardless of the starting width, the water climbs to overflow within a few diurnal cycles; a wider initial gap only delays the approach. The common approach curve reflects that the water level is set by the current effective width alone, not by the crack's original size.

modulated in reach over the cycle, so the ice load it deposits is uneven in depth and time. Translating this uneven, lip-concentrated deposition into a net channel narrowing requires knowing how the ice is redistributed by viscoelastic creep; reconciling the condensation mass budget with the ice convection that carries the deposit away from the lip is the main open problem left for future work.

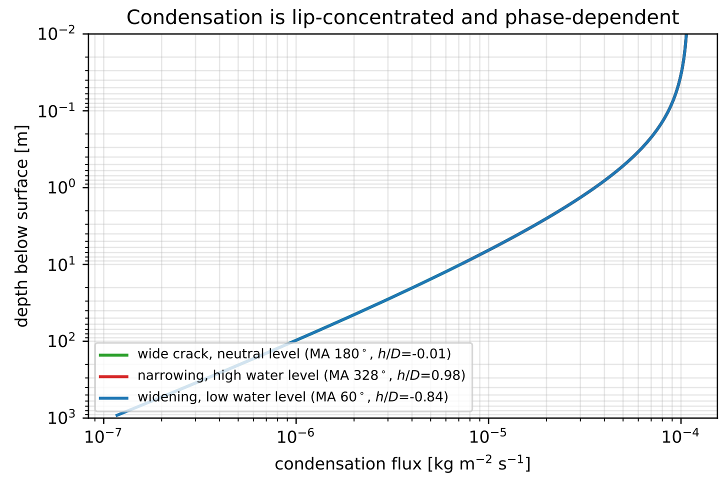

Figure 5. Instantaneous wall condensation rate versus depth below the crack mouth (log–log; equilibrium depth 5 km, Δw = 20 mm, w_{eff} = 8 mm) at three phases of the diurnal cycle: a wide crack at a near-neutral water level (widening phase), a narrowing crack with the water risen to near the surface, and a widening crack with the water drawn down. Condensation concentrates within the thermal skin depth of the mouth at all phases, but the exposed window over which it acts ranges from the topmost few metres (high water) to the full ∼500 m column (low water), marked by the filled circle at each curve's lower end (the water level). The deposition is thus spatially uneven and strongly phase-dependent.

3.3 Predicting the phase and strength of the two mass-flux peaks

Once the geometry is on the attractor, the diurnal mass flux is dominated by gas evaporation $w\,\phi_{top}(w,\, D - h)$; the liquid overflow contributes less than 5%. The way this flux develops as a crack seals is shown in Figure 6. Early in the sealing, when the crack is still wide, the diurnal flux is a single broad pulse peaking near the maximum-width phase (MA ≈ 180°); as condensation narrows the effective width and the water climbs toward the surface, a second maximum emerges on the closing side of the cycle (MA ≈ 290°), where the shortening gas column raises the evaporation rate. The near-close-up curve thus carries the two-peak structure that is the signature of the attractor; the emergence of the late-cycle peak, rather than any change in forcing, is what produces the secondary emission. This flux has two peaks, both with a simple predictor.

The *widening peak* occurs near the maximum-crack-width phase (MA ≈ 180–215°, approaching 180° for stronger forcing) and its strength scales with the throughput there, $w_{\max}\,\phi_{top}(w_{\max}, D - h_{180})$. The *approach peak* occurs near the maximum-water-level phase ψ_h (MA ≈ 270–330°, with the flux maximum lying ∼10–40° earlier than ψ_h because the width is already declining); its strength scales with $w(\psi_h)\,\phi_{top}(w(\psi_h),\, D - h_{\max})$, and because ϕ_{top} rises steeply as the gas column shortens, it grows sharply as the water nears the surface. The two peaks are therefore separated by $\psi_h - 180° \approx$ 90–140°, set by the forcing geometry.

The relative strength is governed by the closeness of approach $h_{\max}/D$ — the same attractor variable of Section 3.2 (Figure 7). For a deep water table the widening peak

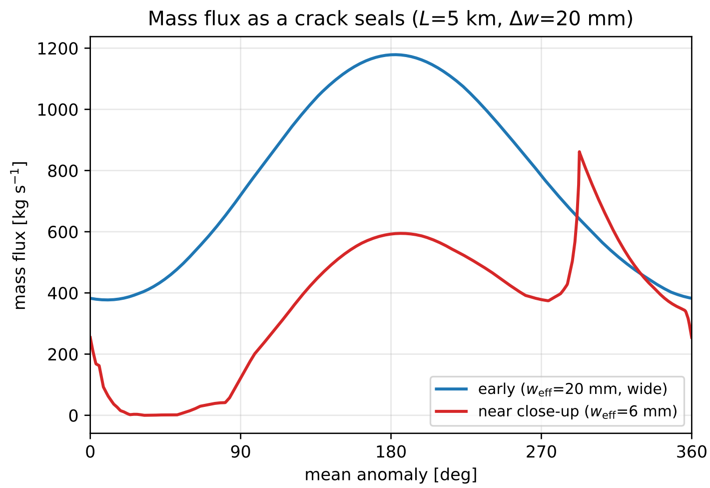

Figure 6. Diurnal mass flux (per-length flux × 500 km) of a sealing crack at two stages (L = 5 km, Δw = 20 mm). Early, while the crack is still wide (w_{eff} = 20 mm), the flux is a single broad pulse near the maximum-width phase. Near close-up (w_{eff} = 6 mm), once the water reaches the surface, a second maximum appears on the closing side of the cycle (MA ≈ 290°) — the approach peak — while the overall throughput drops as the gap narrows. The two-peak profile is a consequence of the sealing, not of the tidal forcing, which is identical for both curves.

dominates and the approach peak is a weak secondary feature (or absent); as the sealing drives $h_{\max}/D \to 1$, the approach peak strengthens and overtakes the widening peak. Thus the attractor not only produces a secondary peak but determines which of the two is the main one. Identifying the stronger (approach) peak with the observed main peak and the weaker (widening) peak with the secondary peak, and allowing the single global phase offset the data permit, the predicted pair maps onto main MA 180° / secondary MA ≈ 50–60°, reproducing the observed phase structure and the ∼130° separation.

3.4 Comparison with the observed emission

The model also reproduces the absolute magnitude of the plume. Multiplying the diurnal mass flux per unit crack length by the ∼500 km summed length of the tiger stripes (Spitale & Porco, 2007), representative attractor cases (sealed to $w_{\mathrm{eff}} \sim$ mm, with the water reaching the surface) give a cycle-mean total emission of ∼230–490 kg/s, in agreement with the 200–400 kg/s inferred from Cassini UVIS solar occultations (Hansen, Shemansky, et al., 2011; Hansen, Esposito, et al., 2017). For the same cases the secondary-to-main peak amplitude ratio is below unity, i.e. a weaker secondary peak, consistent with the observed slab-density profile (Ingersoll, Ewald, et al., 2020). Because the total scales linearly with the assumed active length, this magnitude is an order-of-unity check rather than a tight constraint; conversely, the secondary-to-main ratio depends on the absolute swing Δw and the depth, and could be inverted to constrain them.

The model also reproduces the *phases* of the emission. The diurnal cycle carries a single free parameter that the data cannot fix independently — the absolute orbital phase of the tidal forcing relative to the geometry — so we allow one global phase offset and align the stronger (approach) peak with the observed main peak. The weaker (widening) peak then falls close to the observed secondary peak without further adjustment (Figure 8): the model's two maxima land within, or just beside, the phase bands of the observed main (MA ≈ 160–215°) and secondary (MA ≈ 30–70°) peaks seen in

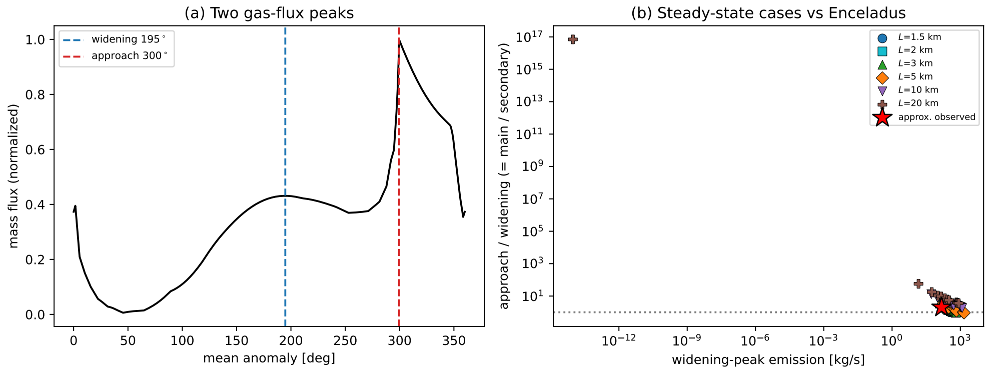

Figure 7. Predictor for the two diurnal mass-flux peaks (equilibrium depth 5 km). (a) The gas-flux curve for a sealed crack ($\Delta w = 10$ mm, $w_{\rm eff} = 7$ mm) showing the widening peak near the max-width phase and the approach peak near the max-water-level phase. (b) Steady-state (sealed, overflowing) cases spanning the tidal swing Δw and the depth L (symbol and colour), on logarithmic axes: the approach-to-widening amplitude ratio — the main/secondary peak ratio — versus the widening-peak emission (per-length flux $\times$ 500 km). The cases span orders of magnitude — deep cracks choke the widening peak and reach large ratios, while shallow cracks keep both peaks and give ratios of order unity — so the approximate observed Enceladus scenario (red star; main/secondary $\approx$2 at an emission of order a few hundred kg/s, an estimate since the absolute secondary-peak flux is not directly measured) lies within the model family.

the slab-density profile (Figure 1; Ingersoll, Ewald, et al. (2020)). The $\sim$100–130° separation between the two model peaks, set by the forcing geometry, thus matches the observed two-peak spacing.

3.5 Fitting the diurnal profile

Beyond matching the peak phases, we fit the forward model directly to the observed diurnal profile of Fig. 1 (slab density versus mean anomaly; Ingersoll, Ewald, et al. (2020)). Placing the crack on the self-sealing attractor fixes the effective width at the overflow seal depth $w^*_{\rm eff}(\Delta w, L)$ — a derived steady-state quantity of the sealing iteration, not a free parameter — leaving as free parameters the absolute tidal swing Δw, the effective source depth L, the second-harmonic amplitude α and phase ϕ_2 of the double-cosine forcing, the along-strike ensemble spread σ_ϕ (defined below), a global phase offset ϕ_0, and an amplitude scale. The amplitude absorbs the unknown slab-density-to-emission proportionality, so only the profile *shape* constrains the physics. We optimise by continuous maximum likelihood (a global differential-evolution search followed by local refinement) and map the parameter posterior with an affine-invariant Markov-chain Monte Carlo (Figure 11).

The best fit (Figure 9) reproduces the main peak, the secondary shoulder, the steep post-peak decline, and the deep trough with a reduced $\chi^2 = 0.78$; a single crack with no ensemble averaging ($\sigma_\phi = 0$) gives $\chi^2 = 2.8$. The posterior medians and 68% credible intervals are $\Delta w = 13.4^{+7.5}_{-2.8}$ mm, $L = 19.2^{+1.9}_{-3.1}$ km, $\alpha = 1.1^{+0.6}_{-0.5}$, $\phi_2 = 93^{+39}_{-27}$ deg, and $\sigma_\phi = 22^{+6}_{-7}$ deg. Two results are robust. First, the source is *deep*: the posterior places L near 19 km with $P(L > 10 \text{ km}) = 0.98$, so the strongly main-peak-dominated profile selects the deep, approach-peak-dominated regime rather than the two-comparable-peak regime of a very shallow crack (Section 3.3). Second, the second-harmonic amplitude is significantly non-zero ($\alpha > 0.3$ at 95% credibility), so the data favour a genuine

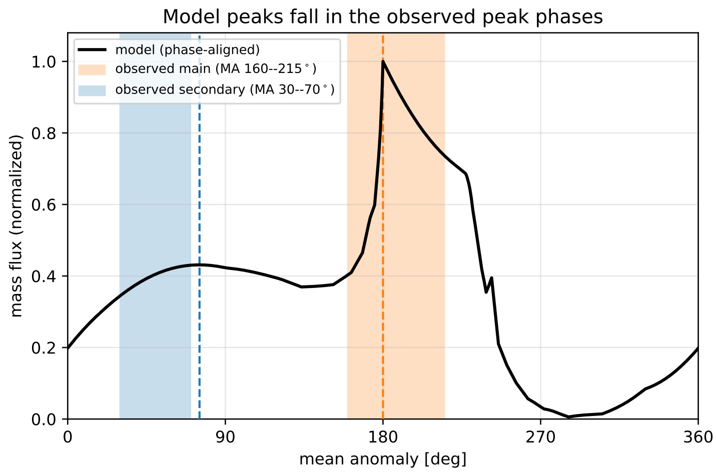

Figure 8. Model diurnal mass flux (normalized; the attractor case of Figure 7a) after applying the single permitted global phase offset that aligns the approach peak with the observed main peak. Shaded bands mark the observed main and secondary peak phases from the Cassini slab-density profile (Ingersoll, Ewald, et al. (2020); Figure 1); no observed flux curve is plotted, only the peak phases. The model's second (widening) peak falls near the observed secondary band without further tuning, and the two model peaks reproduce the observed $\sim 130^\circ$ separation.

double-cosine ($2f$) forcing, consistent with the multi-harmonic character of the tidal stress that drives the walls (Nimmo, Porco, et al., 2014). Two caveats bound the inference: the target is a digitized central-trend curve through a wide, multi-year scatter, and the amplitude is a free scale, so the fit is driven by the profile shape and relative peak heights, not the absolute emission.

[PLACEHOLDER] *Develop the implications of the fitted parameters: interpret the deep effective source depth ($L \approx 19$ km) and reconcile it with the two-peak (shallow) reading of Section 3.3; relate the favoured second-harmonic amplitude to specific tidal-stress models. Add ice-shell-thickness references for the south-polar terrain.*

A single crack produces a sharp, near-cusped approach peak because the piston-driven water reaches the surface rapidly; the model curve in Figure 9 accordingly slightly overshoots the observed peak. The observed slab density, however, is the emission summed over the $\sim$500 km of tiger stripes, whose segments differ in local width, depth, and — through their orientation relative to the tidal-stress field — in forcing phase. Averaging the single-crack profile over this along-strike spread (a periodic Gaussian convolution in mean anomaly of width σ_ϕ) is the physically appropriate comparison to the spatially-integrated data. Fitting σ_ϕ jointly with the crack parameters rounds the peak and markedly improves the agreement, from reduced $\chi^2 = 2.8$ (a single crack, $\sigma_\phi = 0$) to 0.78 once the along-strike ensemble average is included (Figure 10); the ensemble peak amplitude and width then match the observations, including the first (secondary) peak, without further tuning.

4 Conclusions

We modeled the activities of the plumes on Enceladus by treating the crack as a long fissure whose walls widen and narrow over the diurnal cycle. The water level is strongly correlated with the rate of wall motion, and because friction makes the water-level ex-

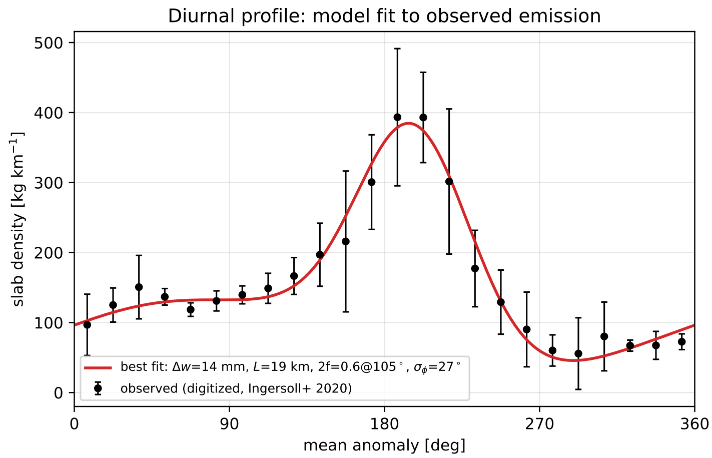

Figure 9. Continuous maximum-likelihood best fit (red) against the digitized observed slab-density profile (black points; error bars are the per-bin scatter spread, not measurement error; Ingersoll, Ewald, et al. (2020), Fig. 1). The crack is on the self-sealing attractor; free parameters are the tidal swing Δw, effective source depth L, the second-harmonic (double-cosine) amplitude and phase, the along-strike ensemble spread σ_ϕ, a single global phase offset, and an overall amplitude (posterior in Figure 11). The single-crack surge that would give a sharp main peak is rounded by the ensemble average over many crack segments (Figure 10).

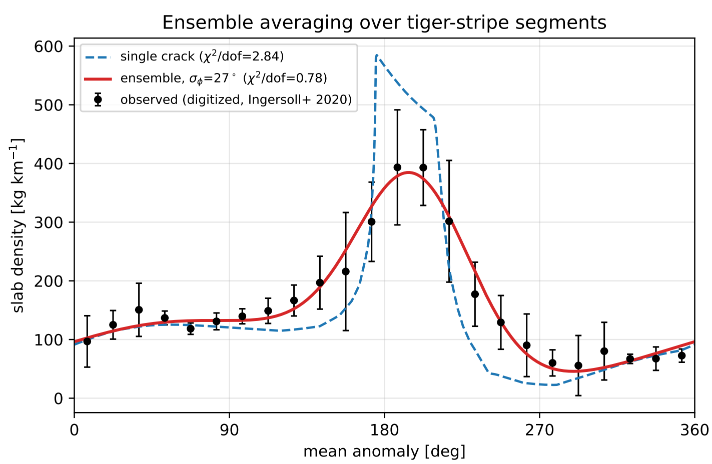

Figure 10. Single-crack versus ensemble-averaged diurnal profile against the observed data. The single crack (blue dashed) surges sharply and overshoots the main peak while missing the secondary; averaging over the along-strike spread of tiger-stripe segments (a Gaussian phase spread σ_ϕ; red) rounds the peak and lifts the shoulder to match the observations — including the first (secondary) peak — improving the reduced χ^2 from 2.8 to 0.78.

cursion scale steeply with the inverse crack width, a narrowing crack drives the water toward the surface.

We noted that an active open crack is a more likely explanation of the observed variability than a closed one, as the warm walls are exposed only when the crack is open.

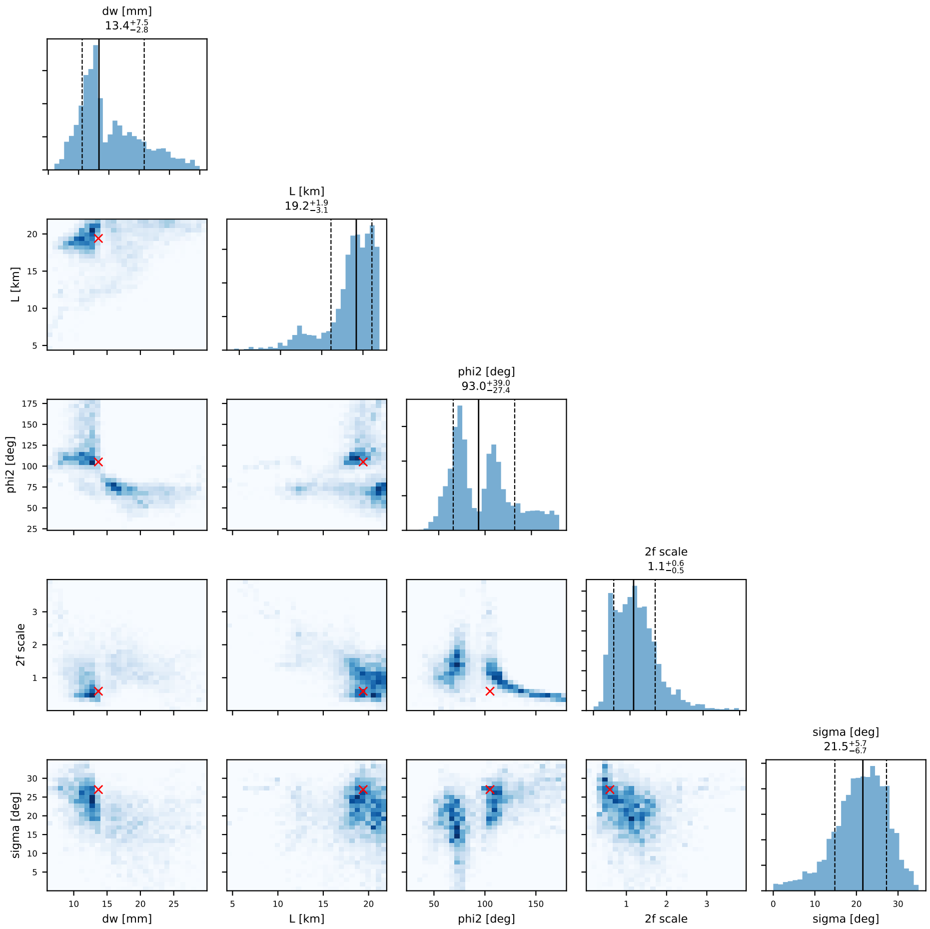

Figure 11. Posterior distribution of the fit parameters from the affine-invariant MCMC (1-D marginals with 16/50/84th percentiles; 2-D joint densities; red crosses mark the maximum-likelihood point). The digitized profile leaves Δw and ϕ_2 loosely constrained, but the depth is firmly deep ($L = 19.2^{+1.9}_{-3.1}$ km, $P(L > 10\text{ km}) = 0.98$) and the second-harmonic amplitude is resolved away from zero ($\alpha = 1.1^{+0.6}_{-0.5}$).

Hence, the proposed model shows an additional evidence that the two-way mass exchange exists between the interior ocean and the planetary surface, which is important for the chemical energy sources for habitability (Parkinson, Liang, et al., 2008).

We further argued that condensation of the plume onto the walls continually seals the crack, and that this self-sealing acts as an attractor: independent of its initial width, any crack that flexes is driven within a few diurnal cycles to a small-width state in which the water reaches the surface and produces the secondary peak. Because the ice lining lowers the mean width but leaves the absolute tidal swing of the walls unchanged, there is no forcing threshold for this overflow state to exist; the swing sets only how far the crack must seal first, and a sufficiently weak swing seals it to a residual gap so small that the plume becomes negligible. The narrow-width overflow regime is therefore a natural end-state rather than a fine-tuned configuration. On this attractor the diurnal mass flux carries two gas-evaporation peaks — a widening peak near the maximum-width phase and an approach peak near the maximum-water-level phase — whose phases and relative strength we predict from the forcing geometry and the closeness of approach; as the water nears the surface the approach peak overtakes the widening peak, and the pair reproduces the observed main / secondary peaks near MA 180° and 50°. The relative strength of the two peaks depends strongly on depth: deep cracks choke the widening peak into a single, high-ratio peak, whereas only shallow cracks retain two comparable peaks. The mere presence of a second peak indicates a source shallow enough to avoid complete choking; a quantitative maximum-likelihood fit of the full diurnal profile with a posterior sampling (Section 3.5), however, is dominated by the single strong main peak and a comparatively weak secondary, and robustly favours a deep, approach-dominated source ($L = 19^{+2}_{-3}$ km, with $P(L > 10\text{ km}) = 0.98$), with the observed secondary emerging once the profile is averaged over the along-strike ensemble of crack segments. Reconciling the modest depth implied by the presence of two peaks with the deeper effective depth from the full-profile fit is left open. How the deposited ice is redistributed from the lip, through fracture and viscoelastic creep, remains an open problem that ultimately sets the long-term balance between sealing and the reopening maintained by the rising water.

While some assumptions are made, the physical processes modeled would follow a similar pattern. Future work can consider a more realistic geometry and generalize the model beyond the current assumptions. Considering an irregular shape of the crack would be possible, but it allows too high a degree of freedom for explorations in the parameter space. Also, the narrowing and widening of the crack may follow a different pattern, where the geometries and viscoelasticity can play a role.

Open Research Section

This is a modelling study and uses no new observational data. The model source code that produces all results and figures in this work — the Python solver (with an optional C++ performance core), the configuration files, and the figure-generating scripts — is openly available in the `EnceladusPlume` repository at `https://github.com/cshsgy/EnceladusPlume`. The observed slab density shown in Figure 1 is reproduced from Ingersoll, Ewald, et al. (2020).

As Applicable – Inclusion in Global Research Statement

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Conflict of Interest declaration

Please include a comprehensive conflict of interest statement that reflect all conflicts of interest for all involved authors. If there are no conflicts of interest please state, "The authors declare there are no conflicts of interest for this manuscript."

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Enter acknowledgments here. This section is to acknowledge funding, thank colleagues, enter any secondary affiliations, and so on.

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